Econ 812 Recitation 7 Notes

Frank Gentile

October 24, 2025

1 Example 1

Question: Stochastic Dynamic Programming [20 points]: Suppose our problem is to

$$\max E_0[\ln(c_0) + \ln(c_1)]$$

by choice of $\{c_0, c_1, k_1, k_2\}$ subject to

$$c_t + k_{t+1} \leq z_t k_t + 0.5 k_t$$

$c_t \geq 0$, $k_{t+1} \geq 0$, and $k_0 = 5 > 0$ given. Suppose $z_0 = 1$ and $z_1 = z^h := 5/4$ with probability $1/2$ and $z_1 = z^l := 3/4$ with probability $1/2$.

Solve this problem using dynamic programming. Identify your state and control variables. Write down the Bellman equations, construct the value functions and optimal policy functions.

Answer: Write out explicitly the constraints:

$$c_0 + k_1 = z_0 k_0 + 0.5 k_0$$

$$c_1 + k_2 = z_1 k_1 + 0.5 k_1$$

With k_0,z_0, and $E_0[z_1]$ given. Also note here that $\beta = 1$. Here the control variable is c_t and state variables are z_t, k_t. The TVC here is that $k_2 = 0$.

Bellman Equation:

$$V(k_0, z_0) = \max_{c_0, c_1, k_1} \log c_0 + E_0[V(k_1, z_1)] \tag{1}$$

Remember the steps to solve a dynamic programming problem:

Step 1 Substitute constraint for c and guess for $V(k')$

Step 2 Solve for policy function

Step 3 Plug policy function back into Bellman

Step 4 Match up coefficients

In this case, since we have $k_2 = 0$, we can say the value of the future state is what you can consume in the future state. Here, the guess for $V(k', z')$ isn't so much of a guess, but rather just the utility of the future state.

$$V(k_0, z_0) = \max_{c_0, c_1, k_1} \log c_0 + E_0[\log c_1] \tag{2}$$

Step 1: Substitute constraints

$$\max_{k_1} \log(-k_1 + z_0 k_0 + 0.5 k_0) + E_0[\log(z_1 k_1 + 0.5 k_1)]$$

Step 2: FOC for k_1

$$\frac{1}{-k_1 + z_0 k_0 + 0.5 k_0} = E_0[\frac{z_1 + 0.5}{(z_1 + 0.5) k_1}]$$

Solve for k_1, with $E_0[z_1] = 1$

$$1.5 k_1 = -1.5 * k_1 + 1.5 * (5 + 2.5)$$

$k_1 = 3.75$. Then $c_0 = 7.5 - 3.75 = 3.75$ and c_1 would depend on the realized value of z_1

2 Example 2

Question: Dynamic Optimization [35 pts]: Consider the following problem:

$$\max \sum_{t=0}^{\infty} \beta^t \log c_t$$

by choice of $\{c_t, k_{t+1}\}_{t=0}^{\infty}$ subject to

$$c_t + k_{t+1} = 2z_t k_t^{2/3}$$

$c_t, k_{t+1} \geq 0$ and $k_0 > 0$ and $\{z_t\}$ given and $\beta = 2/3$.

A. Set up the Lagrangean for this problem, state first-order conditions and the transversality condition (you may ignore the non-negativity constraints on k and c). [7 pts]

B. Eliminate the multiplier on the constraint from your first-order conditions to get a two dimensional first-order difference equation in c and k. [3 pts]

C. Suppose that $z_t = 3/2$ for all t. Solve for the steady state solution $c_t = c > 0$ and $k_t = k > 0$. [10 pts]

D. Log-linearize the non-linear difference equations around the steady state and express the system as

$$A \begin{bmatrix} \hat{k}_{t+1} \\ \hat{c}_{t+1} \end{bmatrix} = B \begin{bmatrix} \hat{k}_t \\ \hat{c}_t \end{bmatrix} + C \begin{bmatrix} \hat{z}_{t+1} \\ \hat{z}_t \end{bmatrix}$$

where A, B and C are 2 x 2 matrices with standard notation

$$\hat{x}_t := \frac{x_t - x}{x}$$

where x is a steady state value for x_t. Use $z = 3/2$ for the steady state value for z_t. [15 pts]

Answer: Steps to solve the lagrangian:

Step 1 Set up Lagrangian, which has form of objective function + constraints*multiplier

Step 2 Take FOCs

Step 3 Combine FOCs to get Euler Equation (usually k_{t+1} and c_t FOC)

Step 1:

$$\max_{c_t, k_{t+1}} \mathscr{L} = \sum_{t=0}^{\infty} \beta^t [\log c_t + \lambda_t (-c_t - k_{t+1} + 2z_t k_t^{2/3})] \tag{3}$$

Step 2:

$$c_t : \frac{1}{c_t} = \lambda_t$$

$$k_{t+1} : \lambda_t = \beta\lambda_{t+1}4/3z_{t+1}k_{t+1}^{-1/3}$$

$$\lambda_t : c_t + k_{t+1} = 2z_t k_t^{2/3}$$

TVC: $\lim_{t\to\infty} \beta^t \lambda_t k_{t+1} = 0$

Part B

Step 3:

EE:

$$\frac{1}{c_t} = \beta\frac{1}{c_{t+1}}4/3z_{t+1}k_{t+1}^{-1/3} \tag{4}$$

RC:

$$c_t + k_{t+1} = 2z_t k_t^{2/3} \tag{5}$$

Part C

From the EE:

$$1 = 2/3 * 4/3 * 3/2 * k^{-1/3} \implies k = (4/3)^3$$

From the RC:

$$c = 2 * 3/2 * (4/3)^2 - (4/3)^3 = 80/27$$

Part D

To log linearize, follow these steps:

Step 1 Take derivative of each side and evaluate at steady state

Step 2 Multiply by steady state variable

Step 3 Multiply by log-linearized variable

Step 4 Continue to next variable

Step 5 Simplify

For the EE:

$$\frac{-1}{c^2} * c * \hat{c}_t = \beta\frac{-1}{c^2}4/3zk^{-1/3} * c * \hat{c}_{t+1} + \beta\frac{1}{c}4/3k^{-1/3} * z * \hat{z}_{t+1} + \beta\frac{1}{c}(-4/9)k^{-4/3}z * k * \hat{k}_{t+1}$$

Simplfying:

$$-\hat{c}_t = -\hat{c}_{t+1} + \hat{z}_{t+1} - 1/3\hat{k}_{t+1}$$

$$\hat{c}_{t+1} + 1/3\hat{k}_{t+1} = \hat{c}_t + \hat{z}_{t+1}$$

And the RC

$$k\hat{k}_{t+1} = 2/3(c+k)\hat{k}_t - c\hat{c}_t + (c+k)\hat{z}_t$$

Arrange into the A, B, and C matrices:

$$A = \begin{bmatrix} 1/3 & 1 \\ k & 0 \end{bmatrix}, B = \begin{bmatrix} 0 & 1 \\ 2/3(c+k) & -1 \end{bmatrix}, C = \begin{bmatrix} 1 & 0 \\ 0 & c+k \end{bmatrix}$$

3 Example 3

Solving stochastic linear difference equations [20 pts]: Suppose we have the following system

$$Et\begin{bmatrix}\hat{k}_{t+1}\\ \hat{c}_{t+1}\end{bmatrix} = \begin{bmatrix}1.5 & -2\\ -3/8 & 1.5\end{bmatrix}\begin{bmatrix}\hat{k}_t\\ \hat{c}_t\end{bmatrix} + \begin{bmatrix}1.5\\ -0.25\end{bmatrix}\hat{z}_t$$

The eigenvalues of W are 0.5 and 2. The corresponding matrix of eigenvectors is

$$P = \begin{bmatrix}2 & -2\\ 1 & 1\end{bmatrix} \text{ with } P^{-1} = \begin{bmatrix}1/4 & 1/2\\ 1/4 & 1/2\end{bmatrix}$$

Suppose that k_t is a pre-determined endogenous variable and c_t is an endogenous jump variable.

A. Does system have a unique bounded solution for any $\hat{k}_0$ and bounded $\{\hat{z}_t\}$? Explain. [5 pts]

B. Suppose $\hat{k}_0 = 0$, and $\hat{z}_{t+1} = (1/2)\hat{z}_t + e_{t+1}$, where $E_t[e_{t+1}] = 0$. Solve for $\hat{c}_t$and k_{t+1} as a function of the state that is part of the unique bounded solution to system above. [15 pts]

Part A

Check the following conditions:

- $n_u = n_y = 1$
- $|\lambda_i| \neq 1 \forall i$
- $R_{uy} = 1/2$ is invertible

So yes the system has a unique bounded solution for any $\hat{k}_0$ and bounded $\{\hat{z}_t\}$

Part B

Two important equations are

$$\begin{bmatrix}s_{t+1}\\ u_{t+1}\end{bmatrix} = J\begin{bmatrix}s_t\\ u_t\end{bmatrix} + P^{-1}A^{-1}C\hat{z}_t \tag{6}$$

and

$$\begin{bmatrix}s_t\\ u_t\end{bmatrix} = P^{-1}\begin{bmatrix}\hat{k}_t\\ \hat{c}_t\end{bmatrix} \tag{7}$$

*Note here that if the order of $x_t = \begin{bmatrix} \hat{k}_t \\ \hat{c}_t \end{bmatrix}$ is switched then this would switch the k_t and c_t, and if the J matrix is stacked unstable then stable, then the s_t and u_t would switch as well. This would also change the R_{uy} portion of the P^{-1} matrix.

Then, to solve this part need to follow these steps:

Step 1: Set up equation 6 to solve for u_t

Step 2: Iterate forward one period

Step 3: Substitute u_{t+1}

Step 4: Find that the part with u_{t+j} drops to 0 since it is multiplied by a number less than 1 raised to an infinite exponent

Step 5: Factor out the exogenous part and substitute law of motion for exogenous shock

Step 6: Set up infinite sum and remember that $\sum_{t=0}^{\infty} \beta^t = \frac{1}{1-\beta}$

Step 7: Plug the result for u_t in for Equation 7.

Step 1:

$$E_t[u_{t+1}] = 2u_t + 1/4\hat{z}_t \implies u_t = 1/2E_t[u_{t+1}] - 1/8\hat{z}_t$$

Step 2:

$$u_{t+1} = 1/2E_{t+1}[u_{t+2}] - 1/8\hat{z}_{t+1}$$

Step 3:

$$u_t = 1/2(1/2E_t[u_{t+2}] - 1/8\hat{z}_{t+1}) - 1/8\hat{z}_t$$

Step 4+5:

$$u_t = -1/8(\hat{z}_t + 1/2E_t[\hat{z}_{t+1}]) = -1/8(\hat{z}_t + 1/2 * 1/2\hat{z}_t + ... + 1/2^j * 1/2^j\hat{z}_t)$$

Step 6:

$$u_t = -1/8\sum_{j=0}^{\infty}(1/2)^j(1/2)^j\hat{z}_t = (-1/8)(4/3) = -1/6\hat{z}_t$$

Step 7:

$$u_t = -1/4k_t + 1/2c_t = -1/6\hat{z}_t \implies c_t = -1/3\hat{z}_t + 1/2\hat{k}_t$$

And from the question we have:

$$\hat{k}_{t+1} = 1.5\hat{k}_t - 2\hat{c}_t + 1.5\hat{z}_t$$

Substituting we have:

$$\hat{k}_{t+1} = 1.5\hat{k}_t - 2(-1/3\hat{z}_t + 1/2\hat{k}_t) + 1.5\hat{z}_t$$

$$\hat{k}_{t+1} = 1/2\hat{k}_t + 13/6\hat{z}_t$$

4 Other useful equations and notes

There are 4 possible combinations of models studied so far:

- Deterministic Finite
- Deterministic Infinite
- Stochastic Finite
- Stochastic Infinite

For each of these combinations you can solve via Lagrangian or Dynamic Programming:

Deterministic

4.1 Finite

Lagrangian

$$\mathscr{L} = \sum_{t=0}^{T} \beta^t [u(c_t) + \lambda_t(-c_t - k_{t+1} + k_t)] + \beta^T \mu_T [k_{T+1} - 0]$$

With TVC: $\beta^T \mu_T k_{T+1} = 0$
DP

$$V_T(k_T) = \max u(c_T) + \beta V_{T+1}(k_{T+1})$$

With $k_{T+1} = 0$ so $V_T(k_T) = u(c_T) \implies c_T^* = k_T$

4.2 Infinite

Lagrangian

$$\mathscr{L} = \sum_{t=0}^{\infty} \beta^t [u(c_t) + \lambda_t[-c_t - k_{t+1} + k_t]]$$

With TVC: $\lim_{t\to\infty} \beta^t \lambda_t k_{t+1} = 0$
DP

$$V(k) = \max_{c_t, k_{t+1}} u(c_t) + \beta V(k_{t+1})$$

Stochastic

4.3 Finite

Lagrangian

$$\mathscr{L} = \sum_{t=0}^{T} \beta^t \sum_{s^t} \Pi(s^t)[u(c(s^t)) + \lambda(s^t)[-c(s^t) - k_{t+1} + z(s^t)k_t]] + \beta^T \sum_{s^t} \Pi(s^t)\mu_T(s^T)k(s^{T+1})$$

DP

$$V(k(s^T), z(s^T)) = \max_{c(s^T), k(s^{T+1})} u(c(s^T)) + E_T[\beta V(k(s^{T+1}), z(s^{T+1}))]$$

And solve backwards

4.4 Infinite

Lagrangian

$$\mathscr{L} = \sum_{t=0}^{T} \beta^t \sum_{s^t} \Pi(s^t)[u(c(s^t)) + \lambda(s^t)[-c(s^t) - k_{t+1} + z(s^t)k_t]]$$

With TVC: $\lim_{j\to\infty} \beta^{t+j}\lambda_{t+j}k_{t+j+1} = 0$ "All dates all states"

DP

$$V(k(s^t), z(s^t)) = \max_{c(s^t), k(s^{t+1})} u(c(s^t)) + E_t[\beta V(k(s^{t+1}), z(s^{t+1}))]$$